

Total No. of Questions :6]

SEAT No :

P167

APR -17/ TE/Insem.-3

[Total No. of Pages :2

T.E. (Civil)

FOUNDATION ENGINEERING

(2012 Pattern) (301009) (Semester-II)

Time : 1 Hour]

[Max. Marks : 30

Instructions to the candidates:

- 1) *Answer Q.1 or Q.2, Q.3 or Q.4 and Q.5 or Q.6 .*
- 2) *Answer to the two sections should be written in separate books.*
- 3) *Figures to the right indicate full marks.*
- 4) *Use of logarithmic tables, slide rule, Mollier charts, electronics pocket calculator and steam tables is allowed.*
- 5) *Assume suitable data if necessary.*
- 6) *Neat diagrams must be drawn wherever necessary.*

Q1) a) Explain the procedure of 'Standard Penetration Test' with a neat sketch. Which are the corrections to be applied ? **[5]**

b) Define

- i) Area ratio
- ii) Inside clearance
- iii) Outside clearance
- iv) R.Q.D.
- v) Recovery ratio

[5]

OR

Q2) a) Determine the depth of overburden above the underlying stiff layer for the following data observed in seismic refraction test. Velocity in upper layer = 580m/s & velocity in lower layer = 4080 m/s. The break in time displacement plot was noted at 30 m/s. **[5]**

b) Explain with a neat sketch (i) Auger boring (ii) Wash boring . **[5]**

P.T.O.

- Q3)** a) Write a note on 'Effect of water table on bearing capacity of soil'. [4]
b) State Terzaghi's equation of bearing capacity for [6]
i) Strip footing
ii) Square footing
iii) Circular footing .
Explain the meaning of each term.

OR

- Q4)** a) The results of two plate load tests on a given location are as follows.
i) diameter = 750mm, settlement = 15 mm, ultimate load = 150kN.
ii) diameter = 300 mm, settlement = 15 mm, ultimate load = 50kN.
Determine the ultimate load on a circular footing of 1.2m diameter causing 15mm settlement. [6]
b) Differentiate between Terzaghi's Bearing Capacity Theory & Meyerhof's Bearing Capacity Theory. [4]

- Q5)** a) Explain with a neat sketch the procedure for determination of pre consolidation pressure. [5]
b) Define the following. [5]
i) Over consolidation ratio
ii) Normally consolidated soil
iii) Coefficient of compressibility
iv) Compression index
v) Coefficient of volume compressibility
vi) Degree of consolidation .

OR

- Q6)** a) Define the following. [4]
i) Differential settlement
ii) Angular distortion
iii) Elastic settlement
iv) Consolidation settlement
b) A 6 m thick saturated clay has a compression index of 0.28. The void ratio at initial stress of 12kN/m² is 2.05. Calculate the settlement and change in void ratio if the stress is increased to 21.6 kN/m² . [6]

